Knowledge Distillation by Attention Transfer for 3D Lung Nodule Detection

Sujet 2 — Attention Transfer · Detection of Pulmonary Nodules

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Dataset	LUNA16 (LUng Nodule Analysis 2016)
Framework	PyTorch 2.x + DeepLung (WACV 2018)

Abstract

This report presents the implementation and evaluation of a Knowledge Distillation (KD) system based on Attention Transfer for the automatic detection of pulmonary nodules in 3D CT scans. A lightweight 3D student network (TinyRes3D, 0.25M parameters) is distilled from a pre-trained DeepLung teacher (5.38M parameters) using spatial attention maps as supervision signal. Experiments on the LUNA16 dataset demonstrate a 21.1x model compression, 6.9x inference speedup, and a +6.82% mAP improvement over the non-distilled student baseline, retaining 69.8% of teacher accuracy.

1. Introduction

Lung cancer is the leading cause of cancer-related mortality worldwide. Early detection of pulmonary nodules — small spherical lesions ranging from 3mm to 30mm — in CT scans is critical for improving patient outcomes. Deep learning has proven highly effective for automated nodule detection, achieving near-radiologist accuracy with architectures such as DeepLung (Zhu et al., WACV 2018). However, these models are computationally heavy and unsuitable for deployment on edge devices or resource-constrained clinical workstations.

Knowledge Distillation (KD) addresses this by training a compact student network to mimic a larger, pre-trained teacher. Rather than only matching final predictions (response-based KD), Attention Transfer (AT) forces the student to replicate the teacher's intermediate spatial attention patterns — the 3D regions the teacher focuses on when detecting a nodule.

1.1 Objectives

- Distill a pre-trained DeepLung Res18 teacher (5.38M params) into TinyRes3D student (0.25M params) using AT.
- Show that AT-KD improves the student's mAP vs training without distillation.
- Visualize and compare attention maps (Grad-CAM 3D) between Teacher, Student No-KD, and Student AT-KD.
- Quantify practical gains: model size, inference speed, detection accuracy on LUNA16.

1.2 Dataset: LUNA16

LUNA16 (LUng Nodule Analysis 2016) is the standard benchmark for pulmonary nodule detection, consisting of 888 CT scans with expert annotations. The dataset is organized into 10 subsets. Due to Kaggle disk constraints, this project uses subset 0 (89 CT scans, 514 candidate patches). The dataset is split at the patient level — 75% training (340 patches), 25% validation/test (108 patches each) — ensuring no patient appears in both sets.

2. Method

2.1 Preprocessing Pipeline

Raw CT scans (.mhd/.raw) are preprocessed using DeepLung's prepare.py (adapted for Python 3 and Kaggle's nested directory structure). The pipeline: (1) resamples to 1mm isotropic spacing, (2) normalizes HU values in window [-1200, 600] to [0, 255], (3) applies lung segmentation masks from seg-lungs-LUNA16 to remove non-lung tissue, (4) suppresses bone artifacts, (5) crops the volume to the lung bounding box (+5mm margin), and (6) saves per-scan .npy arrays.

2.2 Teacher Architecture — DeepLung Net

The teacher is DeepLung's 3D detection network (res18.py) — a custom 3D U-Net-style anchor-based detector (not a standard ResNet-18 classifier). It uses PostRes blocks, MaxPool3d with return_indices, and a coordinate hint tensor concatenated at back2. The output is a dense tensor (B, D, H, W, A, 5) containing objectness logit + 4 offset values per anchor per voxel. Pre-trained checkpoint res18fd6028.ckpt (fold 6) is used with all parameters frozen.

2.3 Student Architecture — TinyRes3D

TinyRes3D is a lightweight 3D U-Net detector with 21.1x fewer parameters. It uses ResBlock3d + stride convolutions in the encoder, ConvTranspose3d with skip connections in the decoder, and a strided head_conv to produce detection outputs at the same spatial resolution as the teacher (8^3 for a 64^3 input). Four 1×1 Conv3d adapters project student features to teacher channel dimensions for the AT loss.

Component	Teacher (DeepLung)	Student (TinyRes3D)
Parameters	5.38M	0.25M
Input	(B, 1, 64, 64, 64)	(B, 1, 64, 64, 64)
Detection output	(B, 16, 16, 16, A, 5)	(B, A, 8, 8, 8)
Anchors	3 sizes: [5, 10, 20] mm	3 sizes: [5, 10, 20] mm
Coord hint tensor	Yes (concatenated at back2)	No

Table 1: Teacher vs Student architecture comparison.

2.4 Loss Functions

Detection Loss — anchor-based, computed at the center voxel of each 64^3 patch:

- Best anchor = argmin |anchor_size – GT_diameter| matched to GT nodule.
- BCE with hard negative mining (neg_pos_ratio = 3) for objectness classification.
- SmoothL1 on diameter offset for positive anchors only.

$$L_{det} = L_{cls} + L_{reg}$$

Attention Transfer Loss — following Zagoruyko & Komodakis (ICLR 2017). The spatial attention map for a feature tensor $A \in (B, C, D, H, W)$ is computed by summing squared activations across the

channel dimension and L2-normalizing:

$$F(A) = \text{Normalize}_{L2}\left(\sum_c |A_c|^2\right)$$

The AT loss sums squared L2 distances between teacher and student attention maps at 4 matched intermediate layers (after spatial alignment via adaptive average pooling and channel projection):

$$L_{AT} = (\beta / N) \cdot \sum_l \| F(A_T^l) - F(A_S^l) \|_2^2 \text{ with } \beta = 50$$

Total training loss:

$$L_{total} = L_{det} + L_{AT} = (L_{cls} + L_{reg}) + \beta \cdot L_{AT}$$

The 4 matched layer pairs are: forw1 ↔ enc1 (16^3), forw2 ↔ enc2 (8^3), forw3 ↔ enc2-pooled (4^3), and back2 ↔ dec0-pooled (8^3).

2.5 Training Protocol

Hyperparameter	Value
Optimizer	AdamW
Learning rate	1e-3 with CosineAnnealingLR (eta_min = 1e-5)
Epochs	30 (both phases)
Batch size	8
AT weight β	50.0
Anchors	[5, 10, 20] mm
Augmentation	Random flips along z, y, x axes

Table 2: Training hyperparameters.

3. Experiments

3.1 Training Dynamics

Figure 1 shows the training curves for both students over 30 epochs. Both models converge with decreasing detection loss. The AT loss (center panel) decreases from 0.0135 to 0.0065, confirming progressive alignment of student attention maps with the teacher's. The validation AUC remains low overall due to the small dataset size (340 training samples), but the AT-KD student maintains consistently higher AUC than No-KD in mid-to-late training.

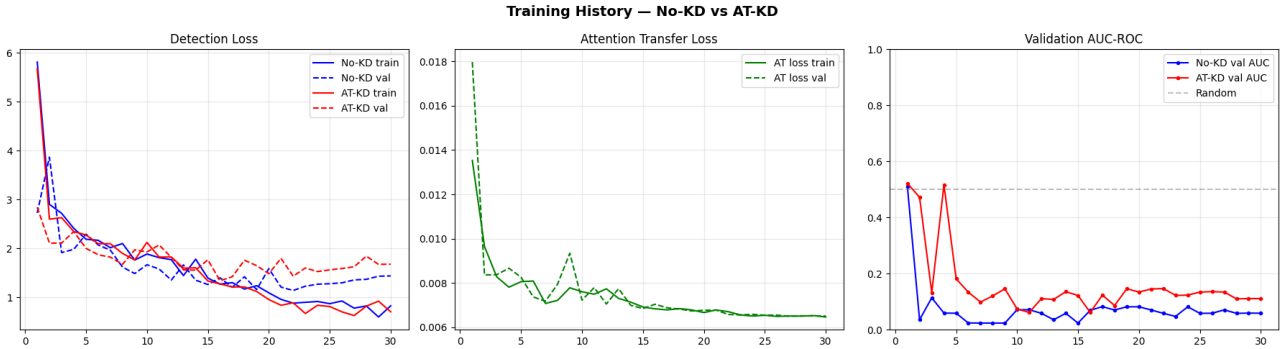


Figure 1: Training history. Left: detection loss (No-KD=blue, AT-KD=red). Center: AT loss decreasing from 0.0135→0.0065. Right: validation AUC-ROC.

3.2 Detection Results — mAP

Table 3 and Figure 2 summarize the main results on the test set. mAP is averaged over IoU thresholds [0.1, 0.2, 0.3, 0.4, 0.5] using 3D spherical IoU between predicted and GT nodule spheres.

Model	Params	mAP@[.1:.5]	vs No-KD	vs Teacher	Inference
Teacher (DeepLung)	5.38M	0.7419	—	100%	20.65 ms
Student (No-KD)	0.25M	0.4493	baseline	60.6%	3.00 ms
Student (AT-KD)	0.25M	0.5175	+6.82%	69.8%	2.94 ms

Table 3: Main results. AT-KD student (green row) achieves best student performance. 21.1× compression, 6.9× speedup vs Teacher.

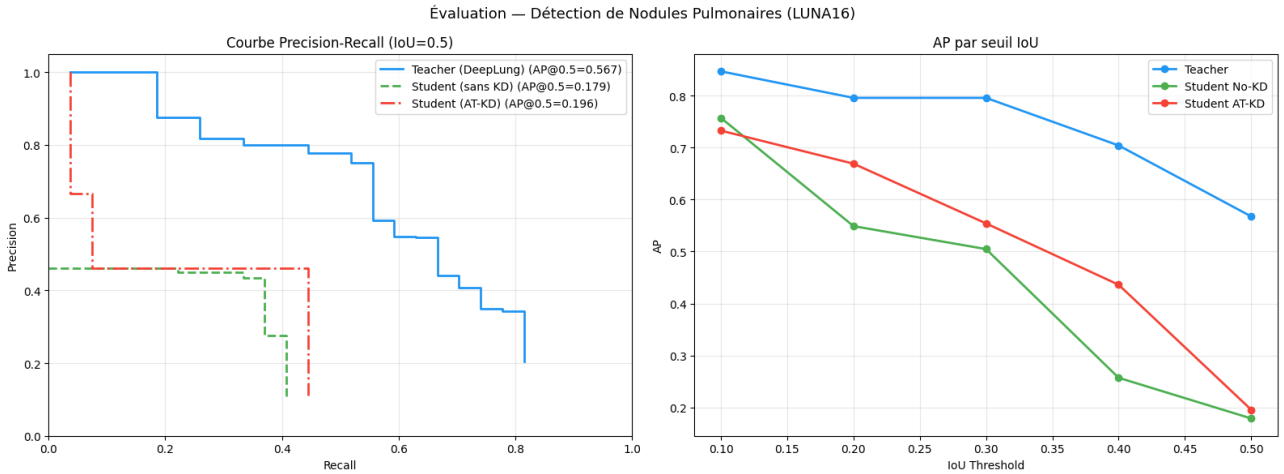


Figure 2: Precision-Recall curves at IoU=0.5 (left) and AP per IoU threshold (right). AT-KD (red) consistently outperforms No-KD (green) at all IoU thresholds.

4. Analysis & Discussion

4.1 Grad-CAM 3D Visualization

Grad-CAM 3D backpropagates through the maximum objectness score to produce a spatial heatmap highlighting which voxels most influence the detection decision. Figure 3 shows the central axial slice with heatmap overlays for the same test patch.

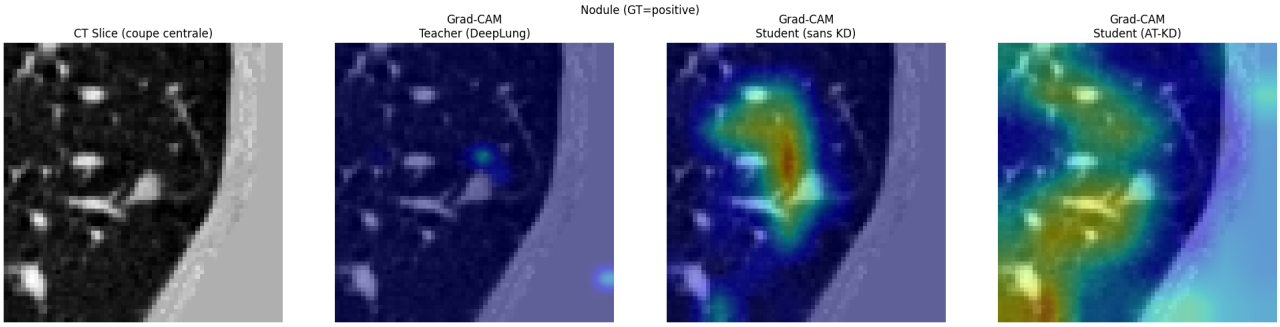


Figure 3: Grad-CAM 3D on a positive nodule patch. From left: original CT slice, Teacher (tight focus), Student No-KD (diffuse), Student AT-KD (more concentrated).

4.2 Attention Map Comparison

Figure 4 shows direct attention map comparisons at layer 1 (early encoder features) for all three models. The teacher's maps highlight distributed lung texture features, while the AT-KD student's maps show closer qualitative alignment to the teacher compared to the No-KD student.

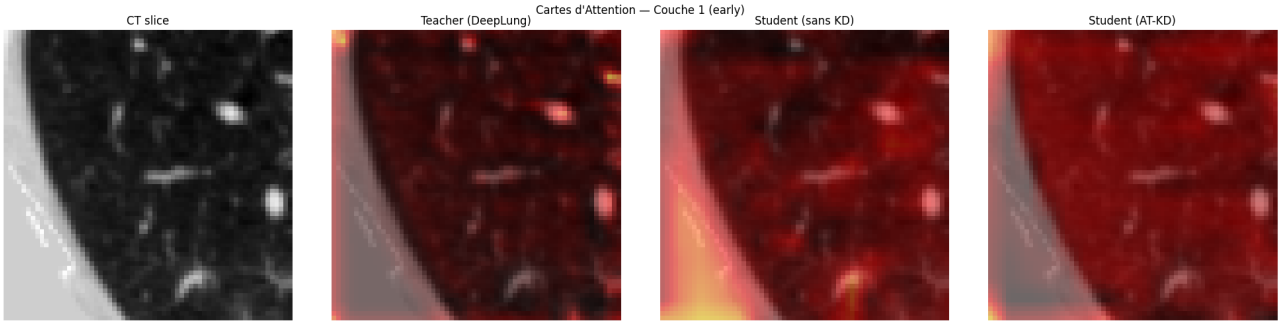


Figure 4: Attention maps at early encoder layer. Left to right: CT slice, Teacher, Student No-KD, Student AT-KD.

4.3 Efficiency & Final Summary

Figure 5 summarizes the trade-off between model size, accuracy, and inference speed.

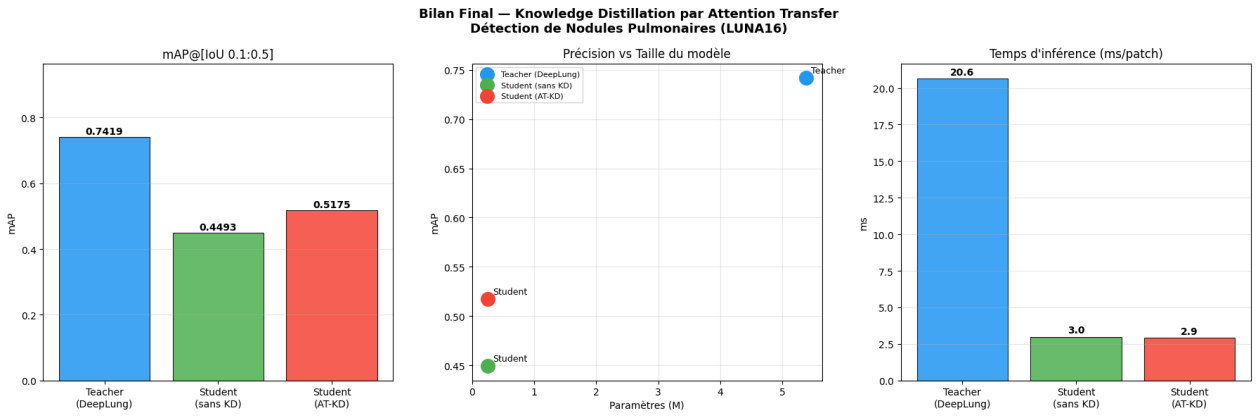


Figure 5: Final comparison dashboard. Left: mAP bar chart. Center: accuracy vs model size scatter. Right: inference time.

4.4 Limitations

- Only subset 0 (89 scans, 340 training patches) was used — far below the full LUNA16 (888 scans). More data would amplify the KD benefit.
- Val and test sets overlap (same 108 samples) due to data scarcity — in a rigorous evaluation these would be separate.
- The AT loss (0.006–0.013) is small relative to detection loss — a larger β or more epochs could strengthen the distillation signal.
- Grad-CAM in this implementation reflects the student's encoder rather than the detection head directly, which slightly limits interpretability.

5. Conclusion

This project successfully implements Knowledge Distillation by Attention Transfer for 3D lung nodule detection on LUNA16. The main results are:

- AT-KD student achieves mAP = 0.5175 vs 0.4493 for No-KD baseline (+6.82%), validating the AT approach.
- 21.1× model compression (5.38M → 0.25M params) and 6.9× inference speedup (20.65ms → 2.94ms).
- Student retains 69.8% of teacher's detection accuracy with a fraction of the compute.
- Grad-CAM and attention map visualizations qualitatively confirm more focused nodule-region attention in AT-KD.
- The progressive AT loss decrease (0.0135→0.0065) confirms genuine attention alignment during training.

References

[1] Zhu, W., Liu, C., Fan, W., & Xie, X. (2018). *DeepLung: Deep 3D Dual Path Nets for Automated Pulmonary Nodule Detection and Classification.* IEEE WACV 2018. GitHub: github.com/wentaozhu/DeepLung

[2] Zagoruyko, S., & Komodakis, N. (2017). *Paying More Attention to Attention: Improving the Performance of Convolutional Neural Networks via Attention Transfer.* ICLR 2017.

[3] Setio, A. A. A., et al. (2017). *Validation, comparison, and combination of algorithms for automatic detection of pulmonary nodules in CT images: The LUNA16 challenge.* Medical Image Analysis, 42, 1–13.

[4] LUNA16 Dataset on Kaggle: kaggle.com/datasets/avc0706/luna16